Expt. 3 γ - Spectrometer (SCA)

(D)

11. Pulse Signals in Nuclear Electronics

As we have seen, modern detectors provide a variety of information on detected radiation in the form of electrical signals. In order to extract this information, however, the signal must be further processed by an electronics system. This system can be designed to perform an enormous variety of tasks. For example, it can be made to sort out the various signals from the detectors, extract energy information, determine the relative timing between two signals, etc., and based on this information, make a decision as to the acceptability of the event. It can then take appropriate action, for example, by activating a recording device. Indeed, many systems virtually run themselves!

In the following chapters, we will try to present an introduction to some of the logic and techniques of setting up an electronics system in nuclear and particle physics experiments and touch on some of the problems which will most likely be encountered. As will be seen, *nuclear electronics* today has been largely standardized into a modular form, i.e., the circuits for the basic processing functions (e.g., amplification, discrimination, etc.) have been built into separate electronic modules of standard mechanical and electrical specifications which are then interconnected as desired. These systems, (NIM, CAMAC), which we will discuss later, are extremely advantageous as they allow the design of many different systems using the same set of modules. Moreover, since the physical and electrical specifications have been standardized and accepted throughout the world, modules from different laboratories can be freely exchanged without any problem in compatibility. An analogous example, perhaps more familiar to the reader, is the modern Hi-Fi system. While the various components (e.g. amplifier, tuner, tape deck, etc.) are constructed separately by many different manufacturers, they are all compatible with each other so that a variety of Hi-Fi systems can be created with no problem.

In both cases, as well, a detailed knowledge of electronics at the level of circuit design is not necessary in order to set up the system, only an understanding of the logic. In the following chapters, therefore, we will assume only an elementary knowledge of circuits and give explanations where some more sophisticated knowledge is needed.

We begin in this chapter with a discussion of pulse signals and their characteristics.

11.1 Pulse Signal Terminology

The coding of information in nuclear electronics is generally done in the form of pulse signals. These are brief surges of current or voltage in which information may be contained in one or more of its characteristics, for example, its polarity, amplitude, shape, its occurrence in time relative to another pulse, or simply its mere presence. This mode of coding, as opposed to other systems, for example, amplitude or frequency modulation of a sinusoidal signal, is natural in nuclear and particle physics, for, as we have

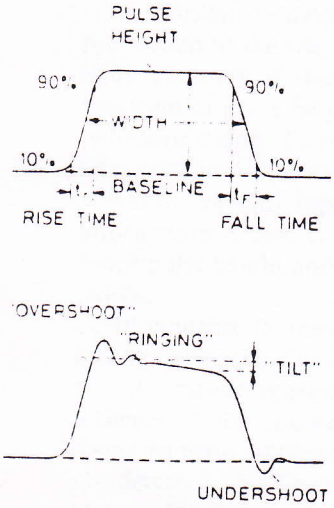

Fig. 11.1. Pulse signal terminology

seen most modern particle detectors are pulse devices. To begin our discussion, let us first identify some basic characteristics of pulse signals and define the terms which are associated with these features. Figure 11.1 (a) shows an ideal rectangular pulse, either in voltage or current, as a function of time. In nuclear electronics, this time scale may vary from micro-seconds to fractions of a nanosecond. We can define the following features:

1) *Baseline.* The baseline of the signal is the voltage or current level to which the pulse decays. While this is usually zero, it is possible for the baseline to be at some other level due to the superimposition of a constant dc voltage or current, or to fluctuations in the pulse shape, count rate etc.
2) *Pulse Height or Amplitude.* The amplitude is the height of the pulse as measured from its maximum value to the instantaneous baseline below this peak.
3) *Signal Width.* This is the full width of the signal usually taken at the half-maximum of the signal (FWHM).
4) *Leading Edge.* The leading edge is that flank of the signal which comes first in time.
5) *Falling Edge.* The falling edge or tail is that flank which is last in time.
6) *Rise Time.* This is the time it takes for the pulse to rise from 10 to 90% of its full amplitude. The rise time essentially determines the rapidity of the signal and is extremely important for timing applications.
7) *Fall Time.* In analogy with rise time, the fall time is the time it takes for the signal to fall from 90 to 10% of its full amplitude.
8) *Unipolar and Bipolar.* Signal pulses may also be unipolar or bipolar. A unipolar pulse is one which has one major lobe entirely (excepting a small possible undershoot) on one side of the baseline. In contrast, bipolar pulses cross the baseline and form a second major lobe of opposite polarity. Figure 11.2 illustrates these two types. Both are used in nuclear electronics.

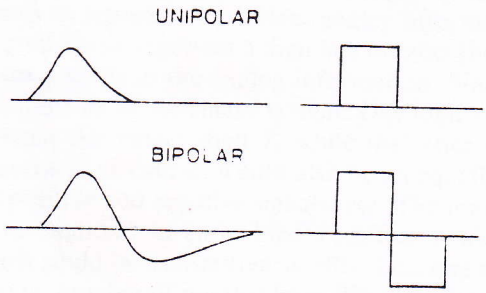

Fig. 11.2. Unipolar and bipolar pulses

In practice, however, it will be found that pulses are very often distorted by various factors in the circuit. Figure 11.1 also illustrates some of various deviations in form which may be observed and the terminology which is used to describe the effects.

11.2 Analog and Digital Signals

Pulse signals, and signals in general, carry information in two forms: *analog* or *digital*. An analog signal codes continuously-valued information by varying one or more of its characteristics, (e.g. amplitude and/or shape), in some fixed relation to the informa-

tion value. The voltage signal from a microphone, for example, *analogically* varies its amplitude continuously in proportion to the intensity of the sound it picks up. Similarly, a scintillation detector, as we have seen, generates pulses whose amplitudes are in proportion to the energy deposited in the detector. If a beam of particles with a continuous spectrum of energies is allowed to strike the detector, then a continuous analog spectrum of pulse heights will result. More generally, if we consider each possible amplitude or shape of a pulse as a *state*, then the analog signal can be said to have an infinite, noncountable number of states. In nuclear physics, most analog pulses are of the amplitude varying type, although in certain cases, such as pulse shape discrimination, information is also contained in the shape of the pulse. Since the proportionality between pulse height and energy is usually linear, these pulses are also referred to as *linear* pulses.

In contrast to the continuum of amplitudes or shapes which are possible for the analog pulse, the *digital* or *logic* signal may only take on a discrete number of states; the information represented is thus of a *quantized* nature. For example, the signal from a Geiger-Müller counter has essentially two states: present or not present.[1] The corresponding information is then simply: *yes*, radiation was detected, or *no*, radiation was not detected. No finer distinction is possible. Similarly, we might imagine a ten-state rectangular signal which can only take on the amplitudes 0, 1, 2, 3, ... 9 V, for example. Such a signal might then be used to represent the decimal integers from 0 to 9, only. Technically, however, it is difficult to find electrical devices which have more than two naturally quantized states. In practice, therefore, all logic signals are limited to two-states only. When reference is made to logic or digital signals, therefore, this generally implies the two-state variety.

Although in a certain sense, the logic signal carries *less* information than the analog signal, from a technical point of view it is the more reliable since the exact amplitude or form of the signal need not be perfectly preserved. Indeed, distortion or noise, which are always present in any circuit, will easily alter the information in an analog signal but would have much less effect on the determination of what state a logic signal is in. Moreover, the limited information-carrying ability of the logic signal may be overcome by using several logic signals to represent equivalent analog information in numerical form, that is, each logic signal would represent a digit in a number (hence the name *digital* signal) whose value corresponds to the analog information. Since only two states are available, this number must be in the binary system. One logic state, for example, *pulse-present* would represent the binary digit *1*, while the other state *pulse-absent* would represent *0*. The contrary, of course, would also be an equally valid system, as would a system involving positive and negative signals, etc. The important point is to keep the same convention throughout. An entire binary number would then consist of a string of logic signals, which could be transmitted serially, i.e., one after the other, or simultaneously along an equal number of parallel lines. This involves, of course, more signals and electronics, however, with the advent of miniaturization and large scale integrated electronics, this disadvantage is removed giving digital systems a clear superiority over analog systems.

In nuclear electronics, the two electrical states of the logic signal are standardized by the NIM convention. One logic state is usually taken as 0 V, i.e., no pulse at all, and the other at a fixed voltage level. Because of the obvious difficulty in generating a pulse

[1] Recall that the Geiger counter signal is saturated so that when it is present, it has the same amplitude and shape regardless of the energy of the radiation detected.

with exactly the right voltage level, a band of voltages into which the signal must fall is defined instead. These limits are given in the next chapter. As a general rule, the analog signals from radiation detectors are changed into logic signals at sometime or another in a nuclear electronics chain. Usually this is performed relatively early after analyzing the detector signal for certain conditions, for example, a certain minimum energy. Here, the detector signal is passed through a *discriminator* which tests the amplitude of the signal for the minimum height. If *yes*, a logic signal is issued, if *no*, no signal (i.e., logic state *no*) is issued. This signal may then be fed into other modules for processing, and depending on the outcome, may initiate further operations: for example, incrementing a counting device or starting a timer, etc. In spectroscopy measurements, where the analog signal is to be analyzed for the exact value of its amplitude, the signal is often *digitized* and the numerical result treated with a computer. Electronic devices which perform such a conversion are know as *analog-to-digital converters* (ADC's). In a similar manner, a digital signal may be converted into an analog signal by a *digital-to-analog converter* (DAC).

As will be seen in Chap. 14, a variety of electronic modules exist for signal processing, some treating analog signals only, others logic signals only, and some converting the two. We emphasize here the importance of distinguishing between the different types. Sending a logic signal into a module expecting an analog pulse or vice-versa does no harm generally, however, unless there is a specific reason for doing so, the result is meaningless.

11.3 Fast and Slow Signals

For technical reasons which we will consider later, it is important to distinguish between *fast* and *slow* pulses in an electronics system. *Fast* signals generally refer to pulses with rise times of a few nanoseconds or *less* while *slow* signals have rise times on the order of hundreds of nanosecond or *greater*. This definition includes both linear and logic signals.

Fast pulses are very important for timing applications and high count rates; in these applications it is very important to preserve their rapid rise times throughout the electronics system. Slow pulses, on the other hand, are generally less susceptible to noise and offer better pulse height information for spectroscopy work.

While it would certainly be more convenient to be able to work with both these types in an identical manner, fast signals, unfortunately, must be treated differently from slow pulses. This is because of their much greater susceptibility to distortion from small, stray capacitances, inductances and resistances in the circuits and interconnections. These elements can combine to form inadvertent, *parasitic* circuits; for example, equivalent RC or RL circuits which have fast transient responses due to the small values of R, C and L (recall that the time constant of the RC circuit is $\tau = RC$). Compared to slow signals these transients are negligibly short. Compared to fast signals, however, these transients are of the same order of magnitude in duration. A fast signal passing through one of these inadvertent circuits can thus be quickly deformed.

A second problem, which will be treated in Chap. 13, is distortion from reflections in the interconnecting cables. This arises because of the short duration of fast pulses relative to their time of transit in the interconnection. Processing fast signals, therefore, not only requires special attention in the circuit design but also in the interconnec-